

Practical No-03: Autoencoder and Variational Autoencoder for Image Reconstruction

MIT Academy of Engineering, Alandi, Pune

Department of CSE AIML

Class T.Y.Tech

Course - Generative AI Lab

Name of Student: Sneha Chaurasia

PRN Number: 202401110046

Batch: A3

Branch: CSE-AIML

Date of Submission: 02/10/2026

GITHUB REPOSITORY URL :

<https://github.com/Sneha529-oss/Autoencoder-and-VAE-for-Image-Reconstruction-Practical-No.-03-Gen-AI>

- **Objective:** Implement an Autoencoder and a Variational Autoencoder (VAE) for reconstructing images and generating new images similar to the training data. Compare the performance of both models based on reconstruction quality, training loss, visual quality of reconstructed images, and quality of newly generated images.
- **Dataset: (Link)**— <http://yann.lecun.com/exdb/mnist/> (MNIST Handwritten Digit Dataset, loaded directly via `tf.keras.datasets.mnist`)

Declaration:

I **Sneha Chaurasia**, confirm that the work submitted in this assignment is my own and has been completed following academic integrity guidelines. The code is uploaded on my GitHub repository account, and the repository link is provided below:

GitHub Repository Link: [<https://github.com/Sneha529-oss/Autoencoder-and-VAE-for-Image-Reconstruction-Practical-No.-03-Gen-AI>]

Signature: Sneha Manoj Chaurasia